Abstract
In this project, I developed a Daily Expense Management System using C++ in Visual Studio. The goal was to create a console-based application that allows users to record, categorize, and view their daily expenses. I used file handling in C++ to store data such as expense categories, amounts, and notes, making the system functional without requiring an external database. The program includes features like category filters, basic budget tracking, and formatted transaction summaries. I structured the code using object-oriented design, which helped me practice using classes, file I/O, and modular functions. Version control was managed using GitHub This project allowed me to apply what I've learned in class to build a practical tool that solves a real-world problem.

Keywords — C++, Personal Finance, File Handling, Visual Studio, Expense Tracker, Console Program

Daily Expense Management System

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Abstract

Many people find it difficult to track daily expenses effectively, resulting in poor budgeting and financial
To address this, I developed a Daily Expense Management System to help users record, categorize, ar
enhancing financial visibility and supporting more efficient budgeting decisions.

1. Project Overview

Many existing financial tools have complex interfaces, lack real-time feedback, and weak categorizatior

- Simple expense recording
- Automatic classification
- Real-time reports and charts
- Alerts for budget control
- Cloud backup and data security

2. Literature Review

- Gupta et al. (2018) emphasized that user-friendly interfaces and visual charts are key to higher adopti
- Chen & Liang (2020) introduced an NLP-based classification method that significantly improved accur
- Singh et al. (2021) highlighted the effectiveness of visual dashboards and budget tracking on user beh
- Kumar & Ray (2019) focused on the importance of encryption and cloud storage for financial data [4].

This project integrates these insights to improve classification, visualization, and security.

5. References

[1] Gupta, R. et al. (2018). Mobile-Based Personal Finance Management System. IEEE.
[2] Chen, Y. & Liang, J. (2020). Expense Categorization Using NLP. Elsevier.
[3] Singh, A. et al. (2021). Budget Monitoring Dashboards. Springer.
[4] Kumar, N. & Ray, S. (2019). Data Security in Finance Apps. IEEE.

3. Methodology

This section outlines the technical approach used in developing the Daily Expense Management System, including tools, structure, and implementation strategies.

3.1 Development Tools and Environment
The project was developed in Python 3.10, using Visual Studio Code as the primary IDE. Version control was managed using GitHub. The program runs on standard Windows and macOS machines without requiring special hardware.

3.2 Data Storage and Management
All expense data is stored in a local SQLite3 database, which is lightweight and easy to integrate with Python. Tables were created for user accounts, transactions, categories, and budget limits. The database enables CRUD operations, data persistence, and category-based filtering.

Sample schema snippet:

```
CREATE TABLE transactions (
   id INTEGER PRIMARY KEY AUTOINCREMENT,
   date TEXT,
   category TEXT,
   amount REAL,
   note TEXT
);
```

3.3 Program Features and Code Structure
The project uses a modular Python structure with clearly separated files for database operations, interface logic, and chart rendering. Main features include:
- Adding and editing expense records
- Auto-categorization based on keywords
- Generating monthly reports and pie charts
- Budget threshold alerts

Charting is implemented using matplotlib, and the user interface uses Tkinter.

3.4 Security and Backup
The database can be exported for cloud backup, and user data is stored locally with optional encryption. Future versions may integrate Firebase or Google Drive for real-time syncing.

4. Conclusion

Through this project, I was able to turn a real-world problem into a working C++ application. The Daily Expense Management System I created gives users the ability to input their expenses, organize them by category, and view summaries of their spending. By saving the data to text files, I kept the system lightweight and easy to use without extra dependencies.

This experience helped me improve my C++ skills, especially in writing modular code, using file I/O, and designing class structures. Working in Visual Studio gave me a better understanding of project setup and debugging I also practiced using GitHub to save different versions of my code and track changes. Overall, this project gave me confidence in applying programming to everyday needs, and I look forward to improving it by adding features like search, export to CSV, and maybe even a graphical version in the future.